

SECV 3213 ASSIGNMENT 3

COURSE PROJECT PROPOSAL

SEM1-20252026

SECTION : 02

Group members:

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Introduction

This project proposes the development of an image processing application using Python and OpenCV for the course Fundamental of Image Processing. The application is intend to function as a simple image editor that allows the users to upload, edit and save images through an interactive graphical user interface. It will include basic editing controls such as undo, redo and reset as well as image processing features such as image adjustment, effects, cropping and overlays. The goal of this project is to apply fundamental image processing technique in a practical and user-friendly environment.

The Proposed Tasks in the Apps

1. Basic Features

- **Upload and Save Image** - Allows users to import images from their device and save the edited results after processing.
- **Undo, Redo, and Reset** - Enables users to revert or reapply editing actions, and reset the image to its original state at any time for flexible editing control.

2. Crop Menu

- **Freeform Crop** - Allows users to manually enter width and height values for custom cropping.
- **Fixed Size Crop** - Provides predefined cropping ratios, such as square, 1:1, 16:9, and 5:4 for consistent image formatting.

The Proposed Tasks in the Apps

3. Overlay Menu

- **Add Overlay Image** - Enables users to add the cutout image or a second image onto the original image as an overlay.
- **Cutout** - Isolates the selected region from the image by removing the background, preparing it for overlay by choosing Region Selection, Auto Selection, or Outline Selection.
- **Resize and Position** - Allows users to resize and reposition the overlay image to achieve proper alignment.
- **Layer Management** - Provides control over image layers, allowing users to move layers up or down, duplicate layers, or delete unwanted overlays.

The Proposed Tasks in the Apps

4. Effects Menu

- **Noise** - Adds controlled Gaussian noise to the image, where users can adjust the noise size and intensity to achieve the desired visual effect using slider.
- **Grayscale** - Converts the image from RGB color space into grayscale by removing color information while preserving intensity. The effect is applied instantly upon clicking the selection button.
- **Halftone** - Produces a stylized print-like appearance by converting image tones into dot-based or intensity-pattern representations, with adjustable intensity to control the halftone level using slider.
- **Blurring** - Smooths the image by reducing sharp pixel variations using median filtering. The blurring strength can be adjusted to control the level of smoothness using slider.

The Proposed Tasks in the Apps

5. Adjust Menu

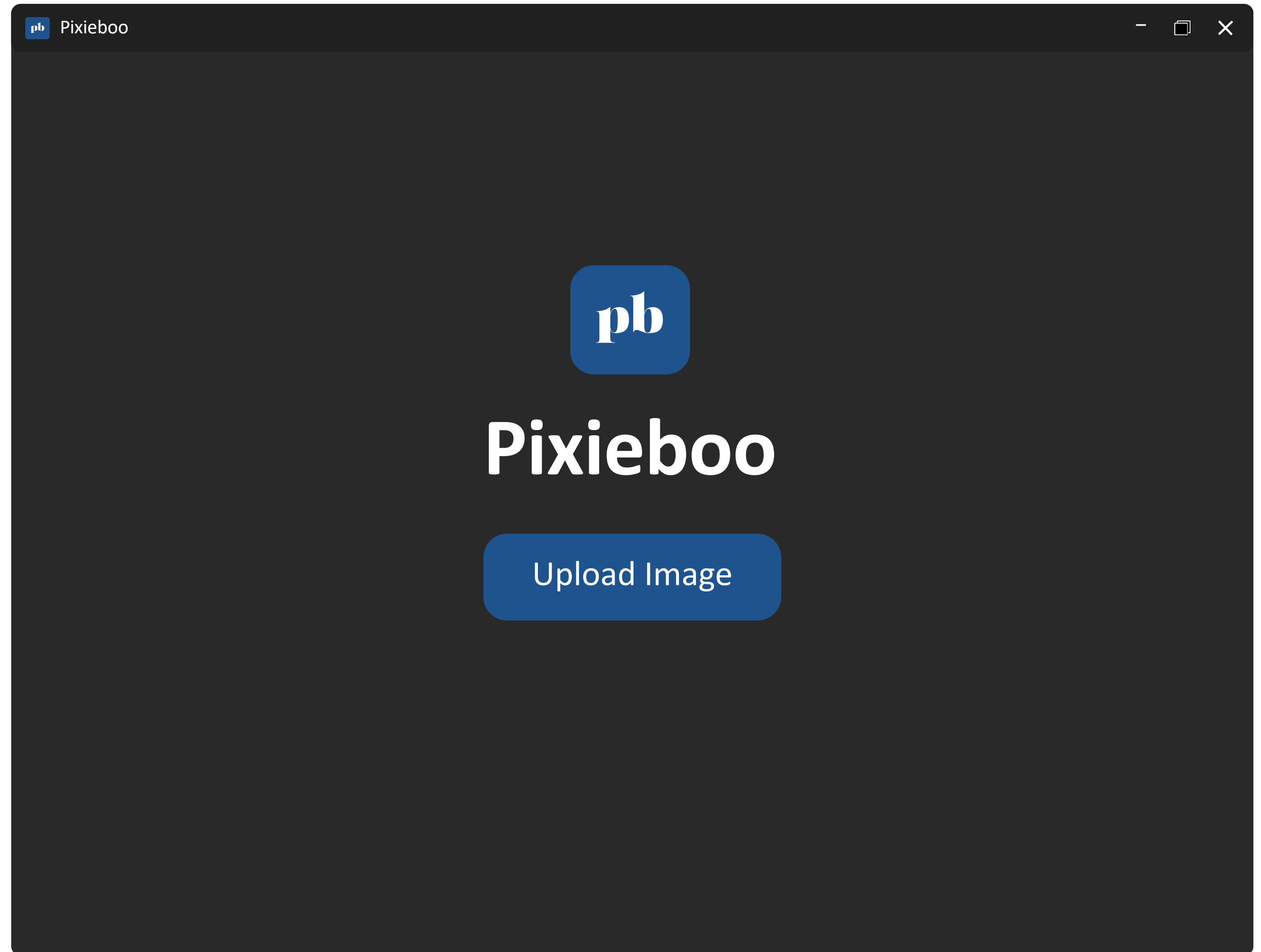
- **Sharpness** - Improves image clarity by enhancing edge details using a Laplacian-based approach that highlights intensity variations. Users can interactively adjust the sharpness level to control the strength of enhancement using an adjustable slider.
- **Brightness** - Modifies overall image brightness by adjusting pixel intensity values through addition operations. The level of brightness can be increased or decreased based on user input using an adjustable slider.
- **Darkness** - Reduces image brightness by scaling pixel intensity values using a multiplication operation. The level of darkness can be increased or decreased based on user input using an adjustable slider.

The Proposed Tasks in the Apps

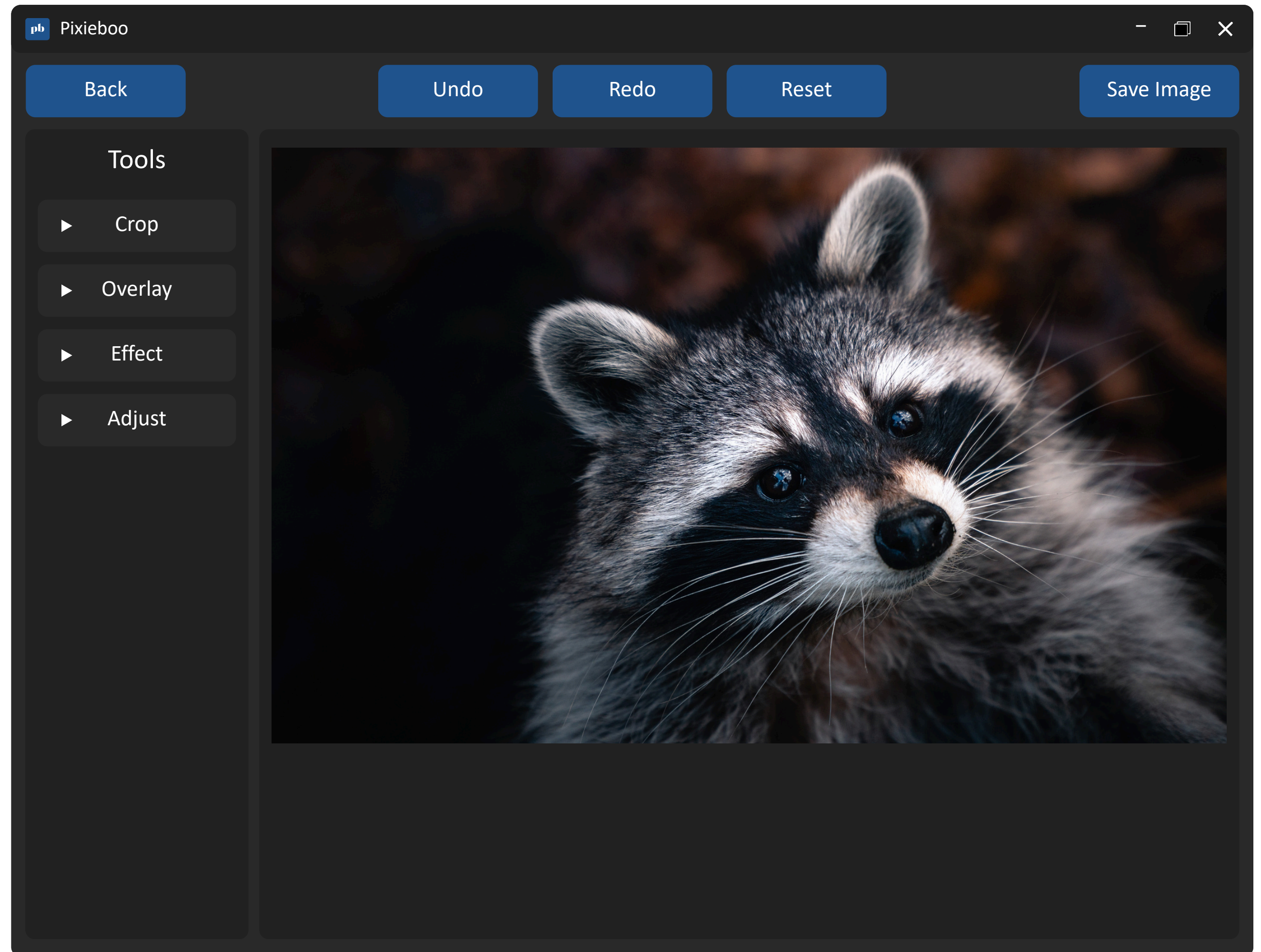
5. Adjust Menu

- **Vignette** - Creates a darkening or lightening effect around the image borders by gradually adjusting intensity values toward the edges. The level of darkness or lightness can be increased or decreased based on user input using an adjustable slider.
- **Denoise** - Removes unwanted noise while preserving important details by combining median filtering, Gaussian smoothing, and high-boost filtering. The effect is applied instantly upon clicking the selection button.

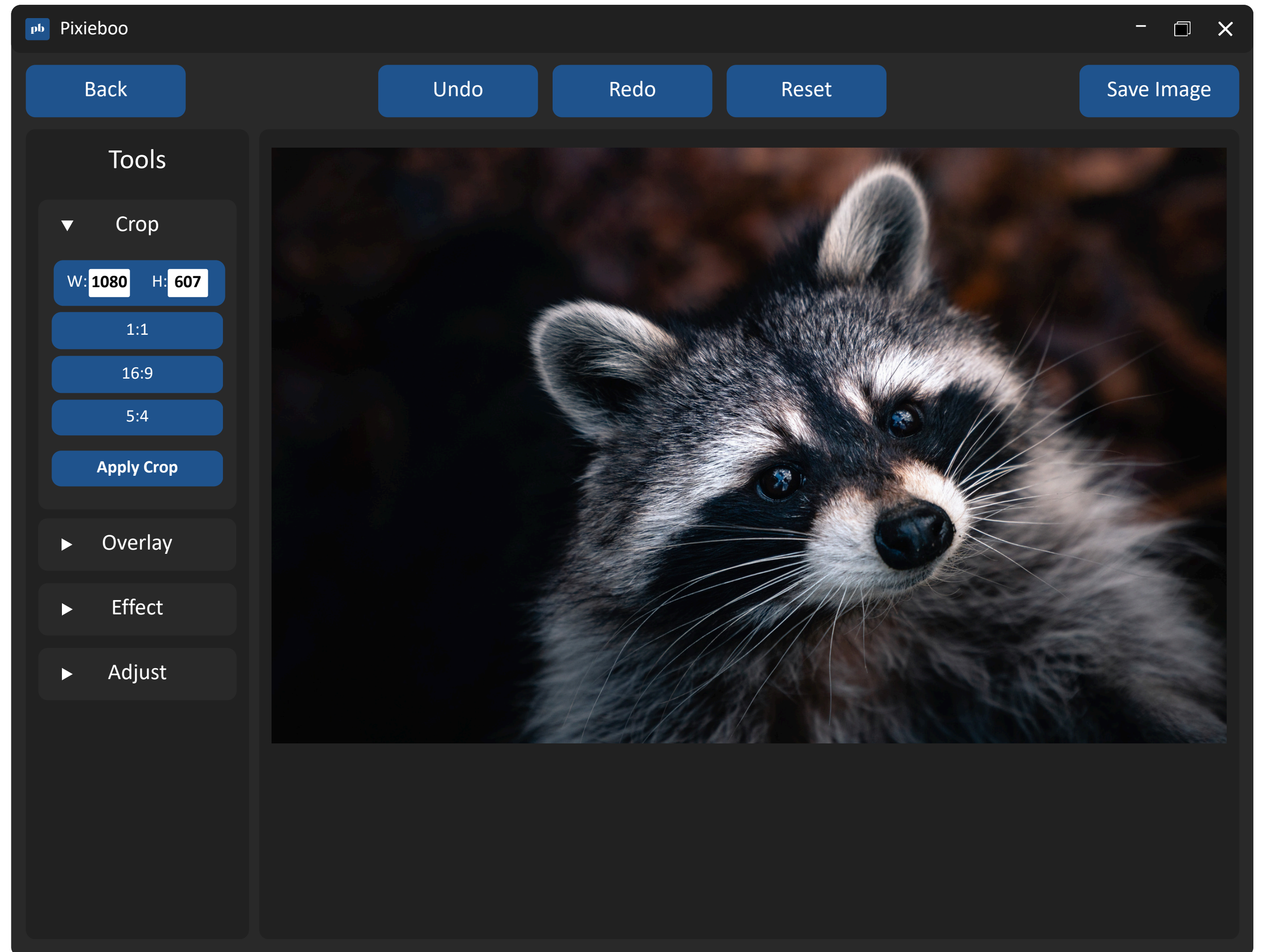
The Proposed Interface & Interaction



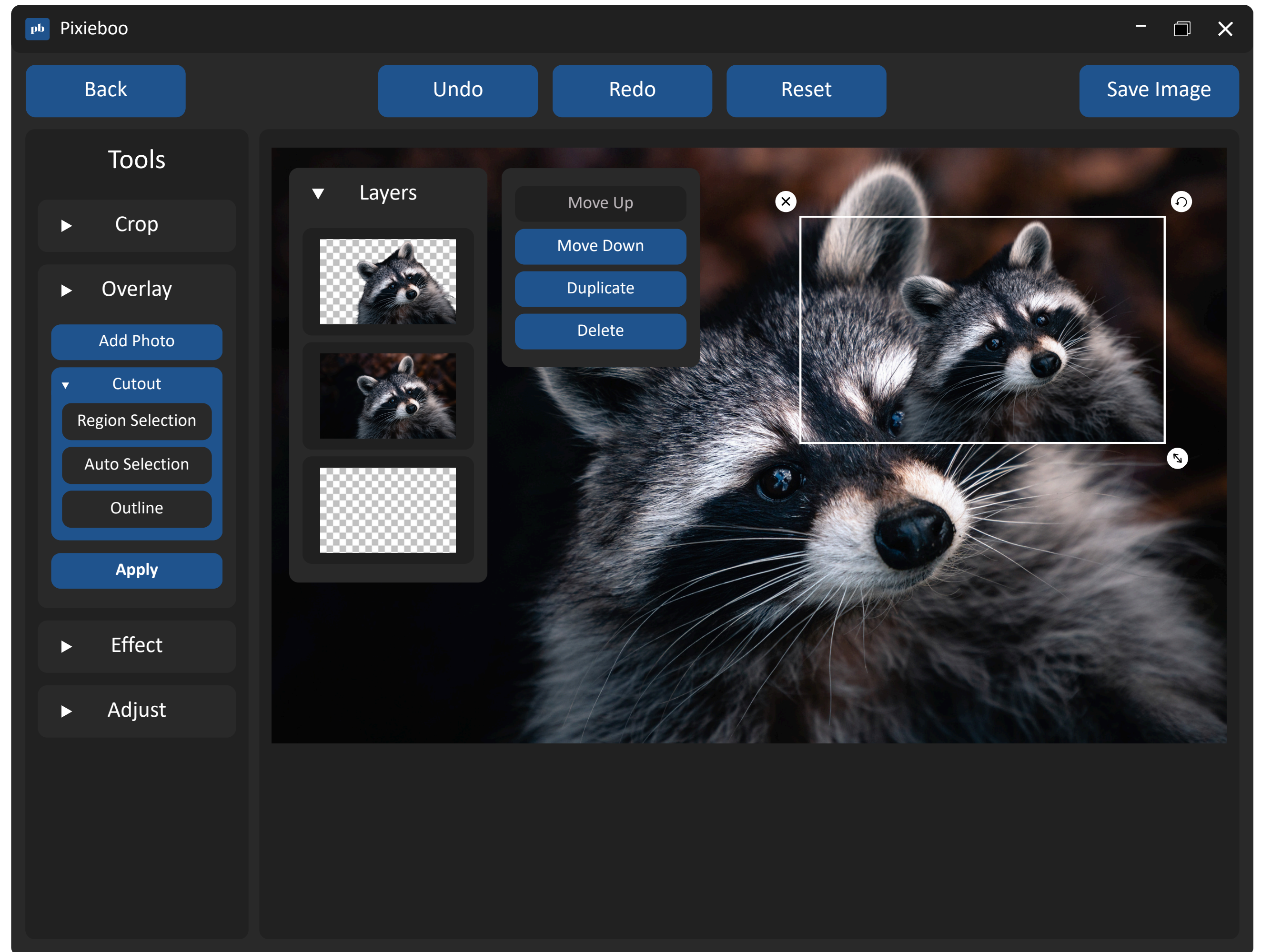
The Proposed Interface & Interaction



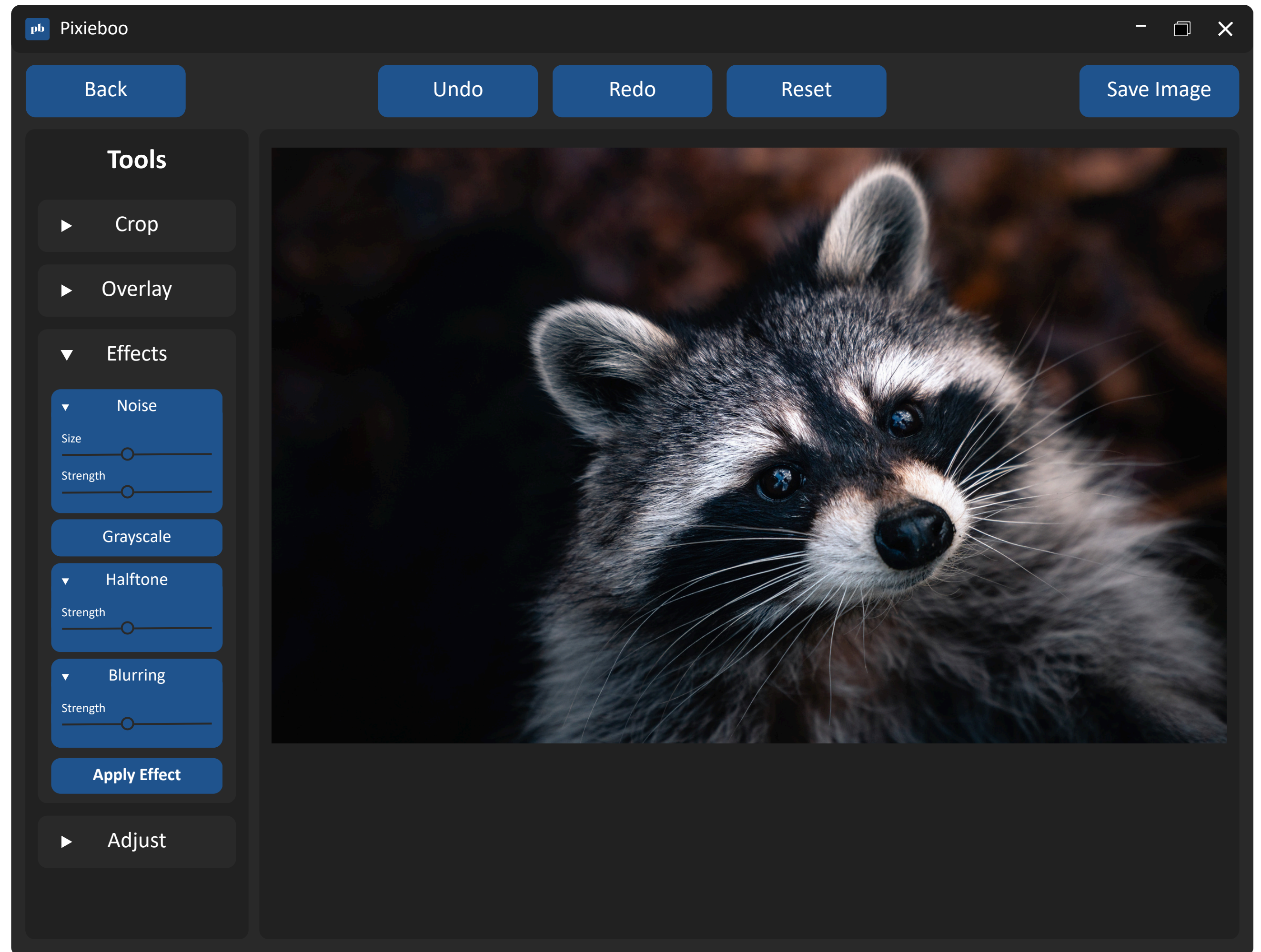
The Proposed Interface & Interaction



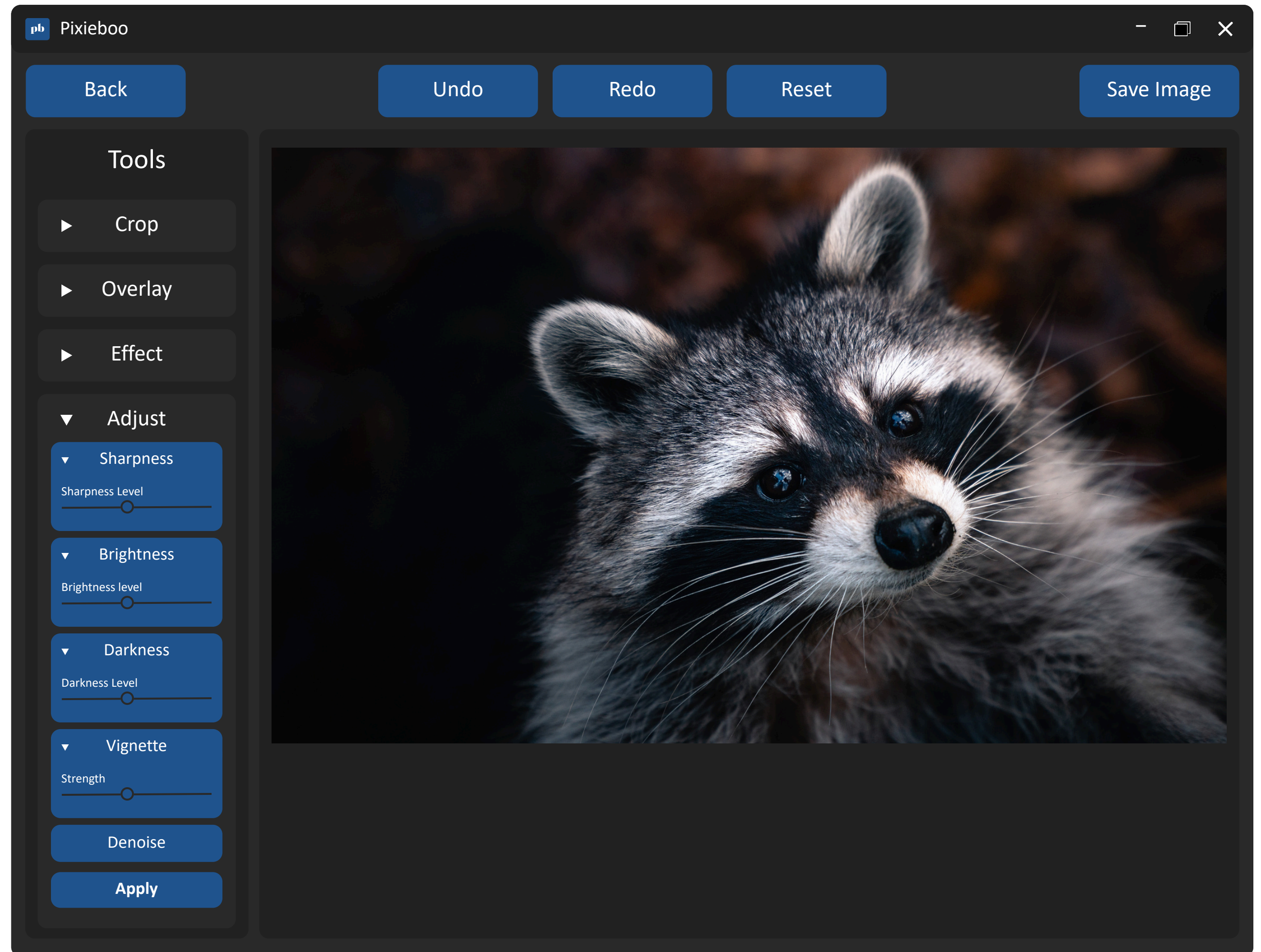
The Proposed Interface & Interaction



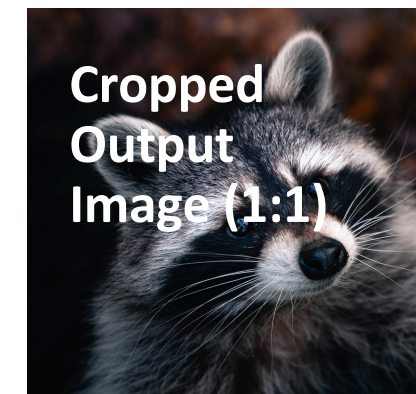
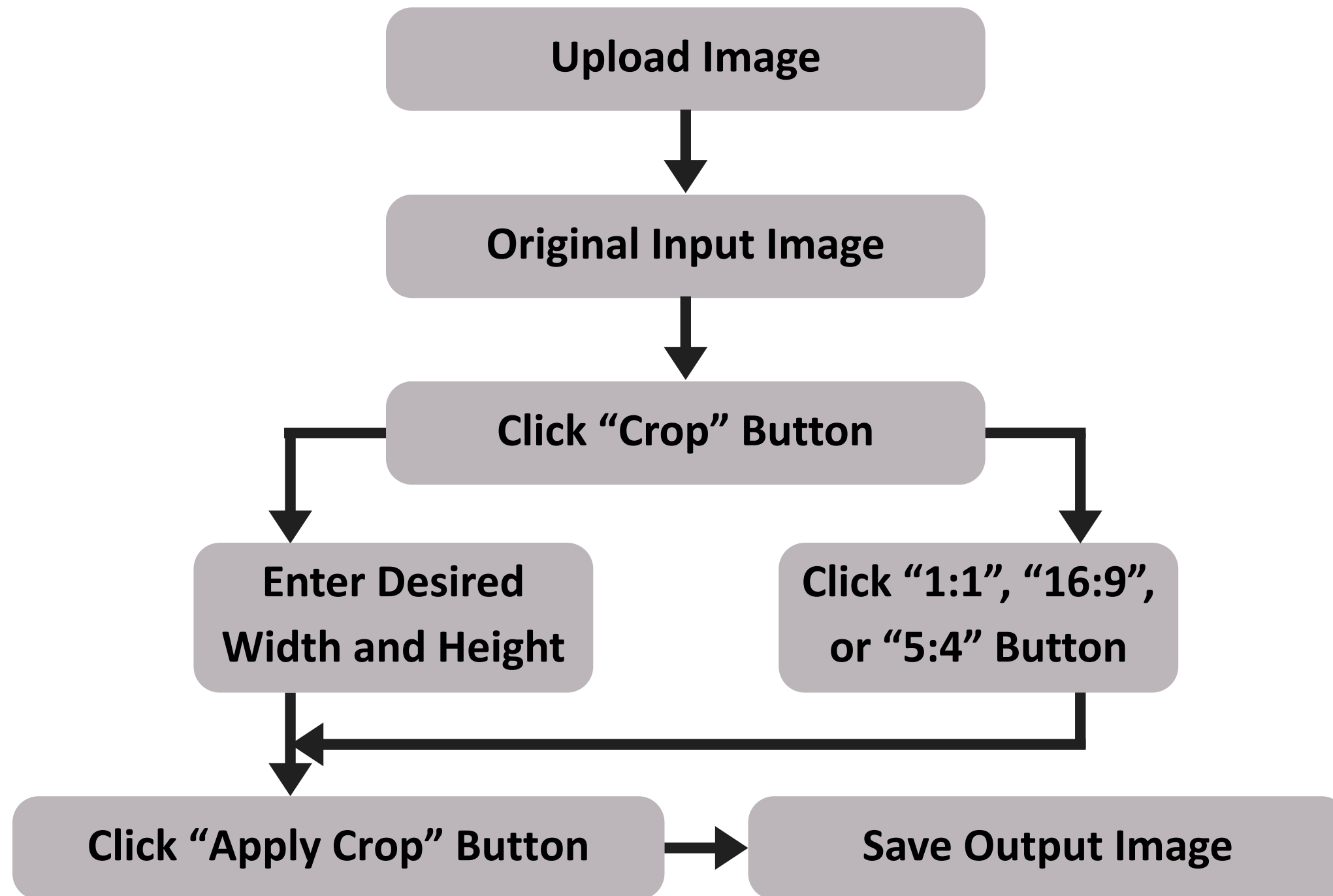
The Proposed Interface & Interaction



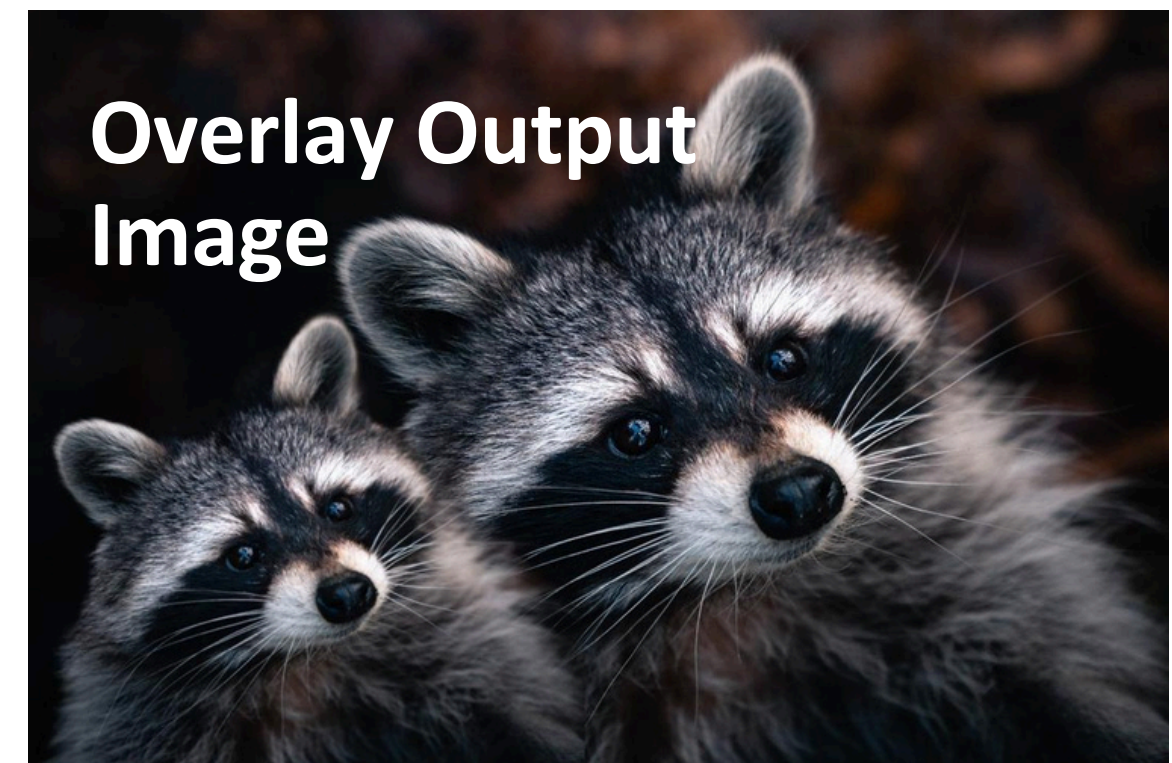
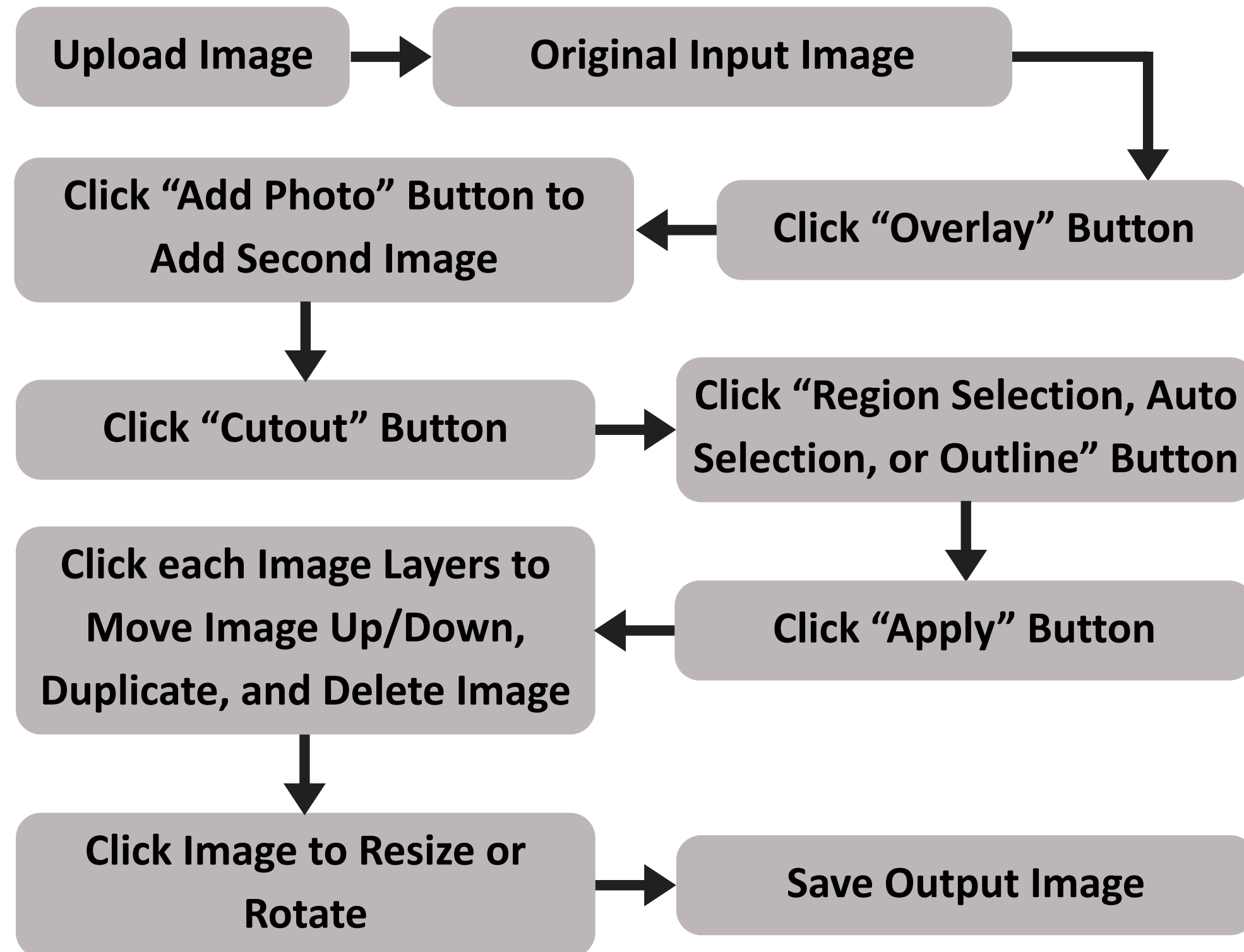
The Proposed Interface & Interaction



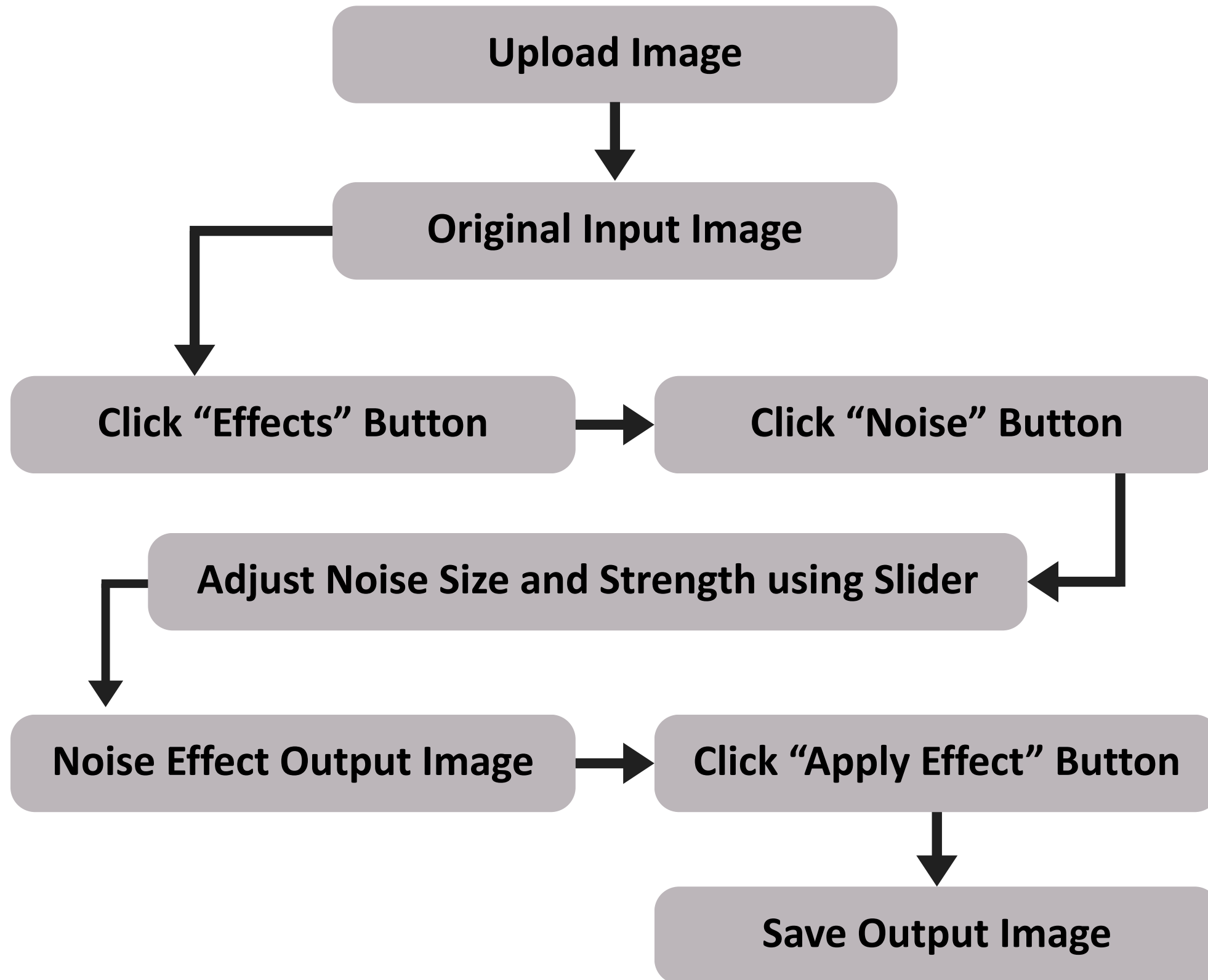
Sample Input and Output Process (Crop)



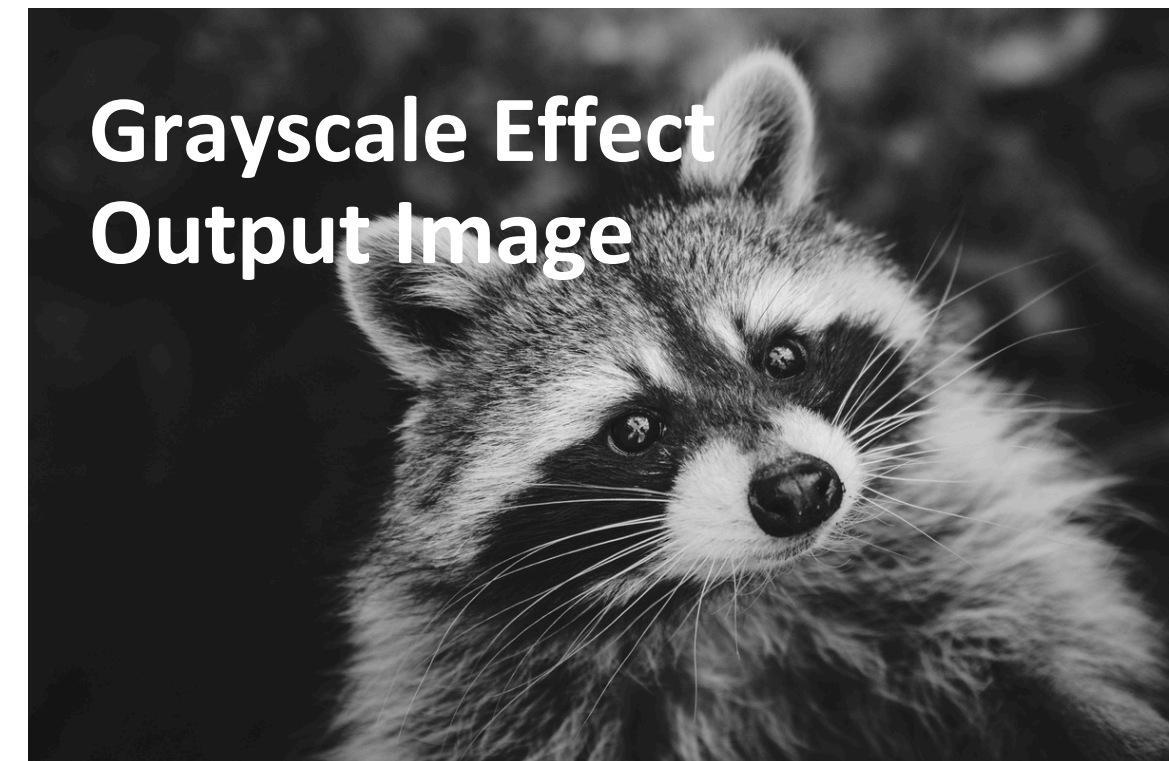
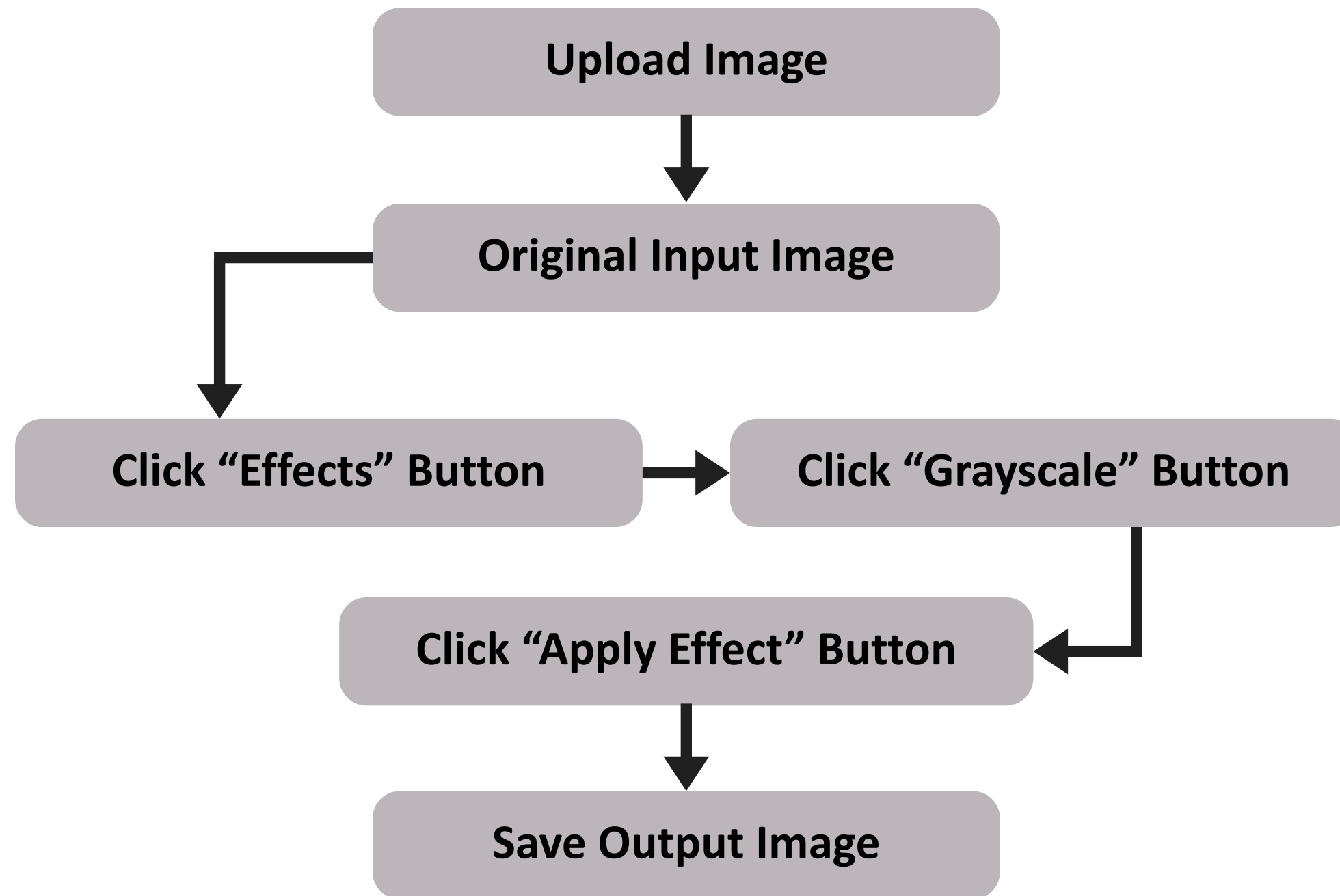
Sample Input and Output Process (Overlay)



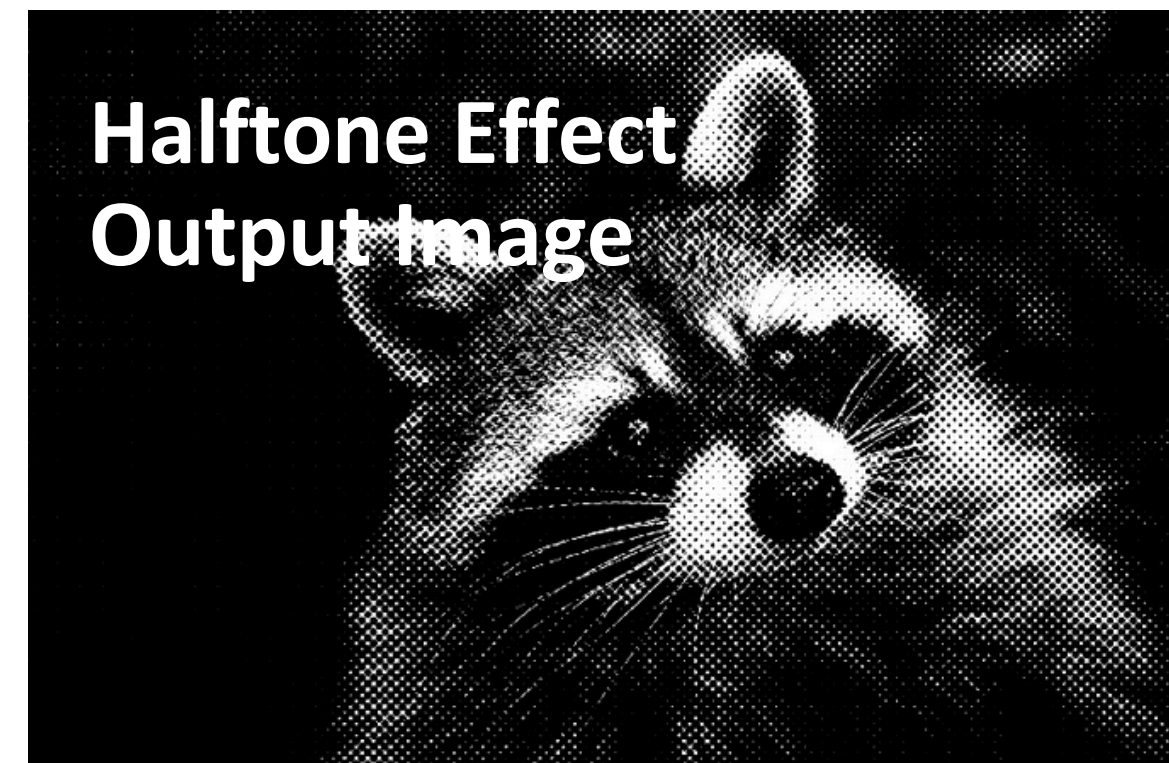
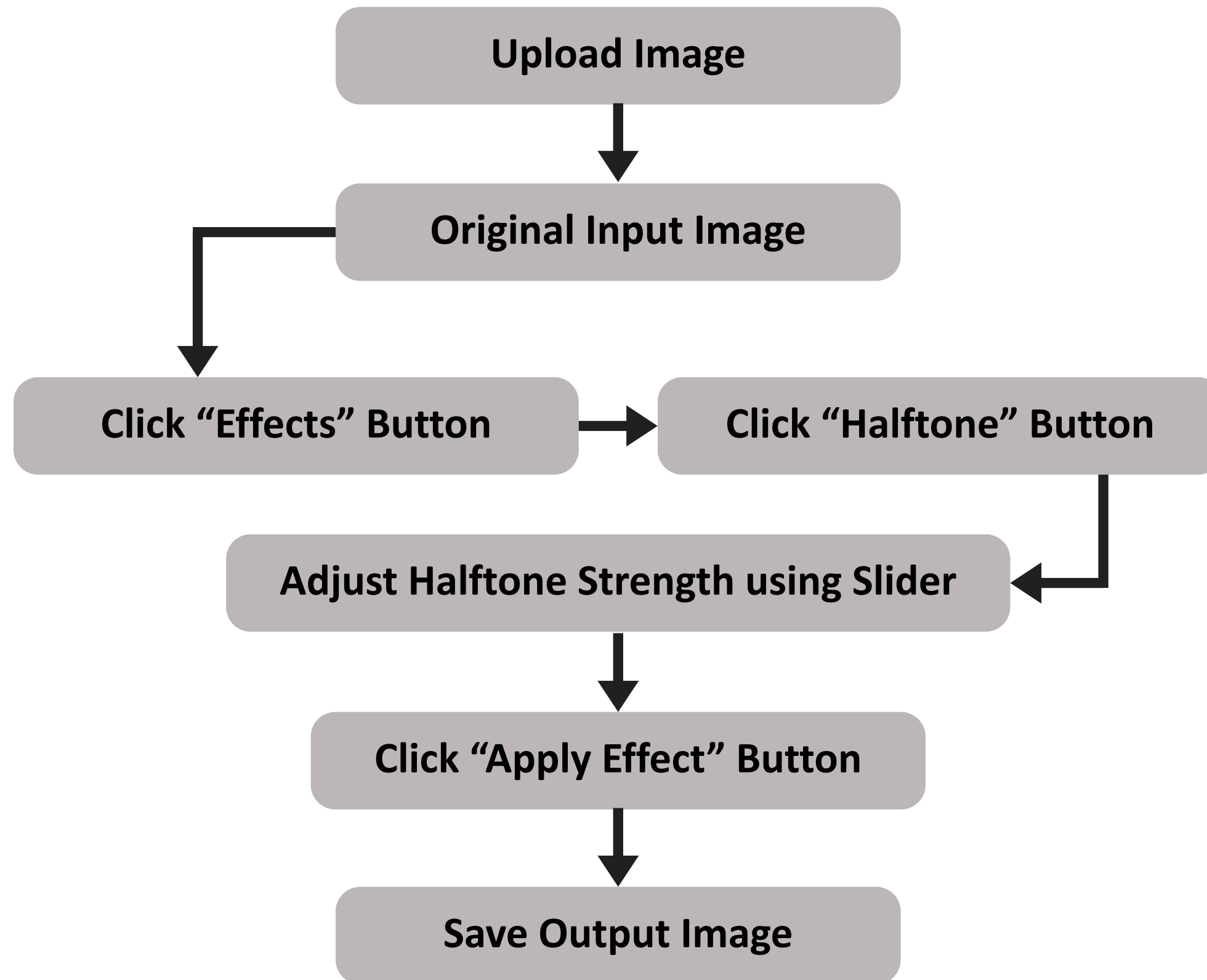
Sample Input and Output Process (Noise)



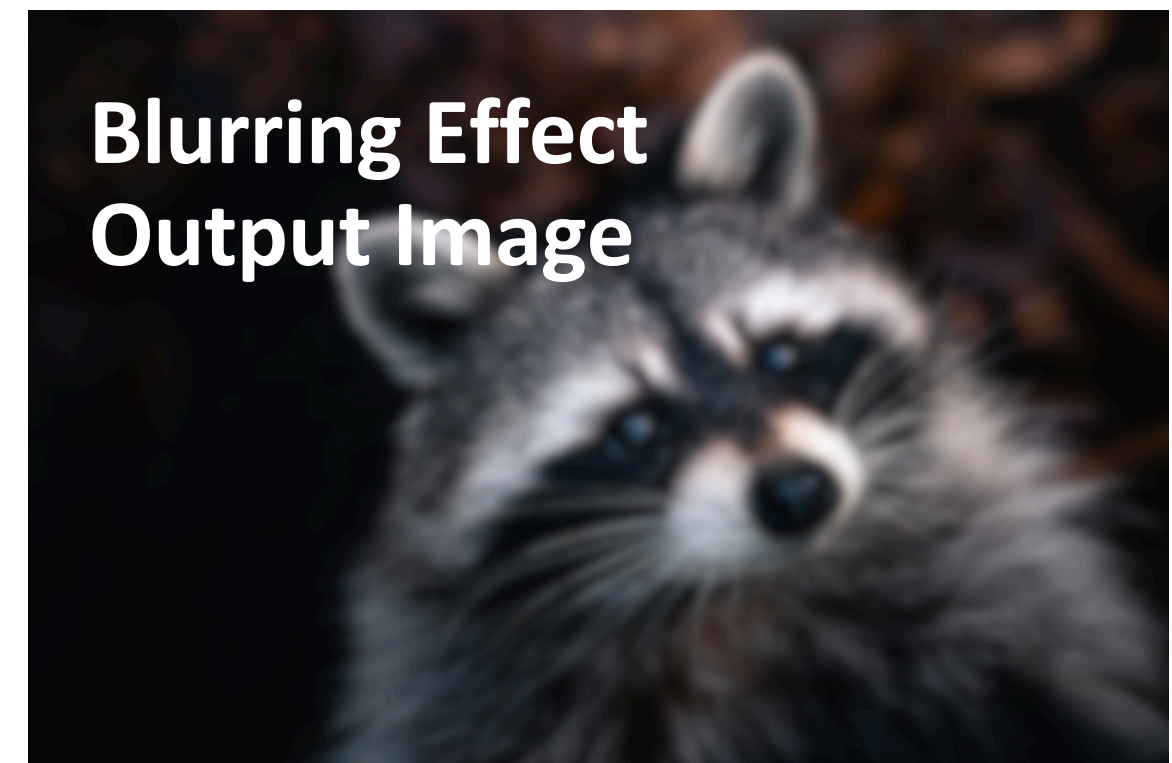
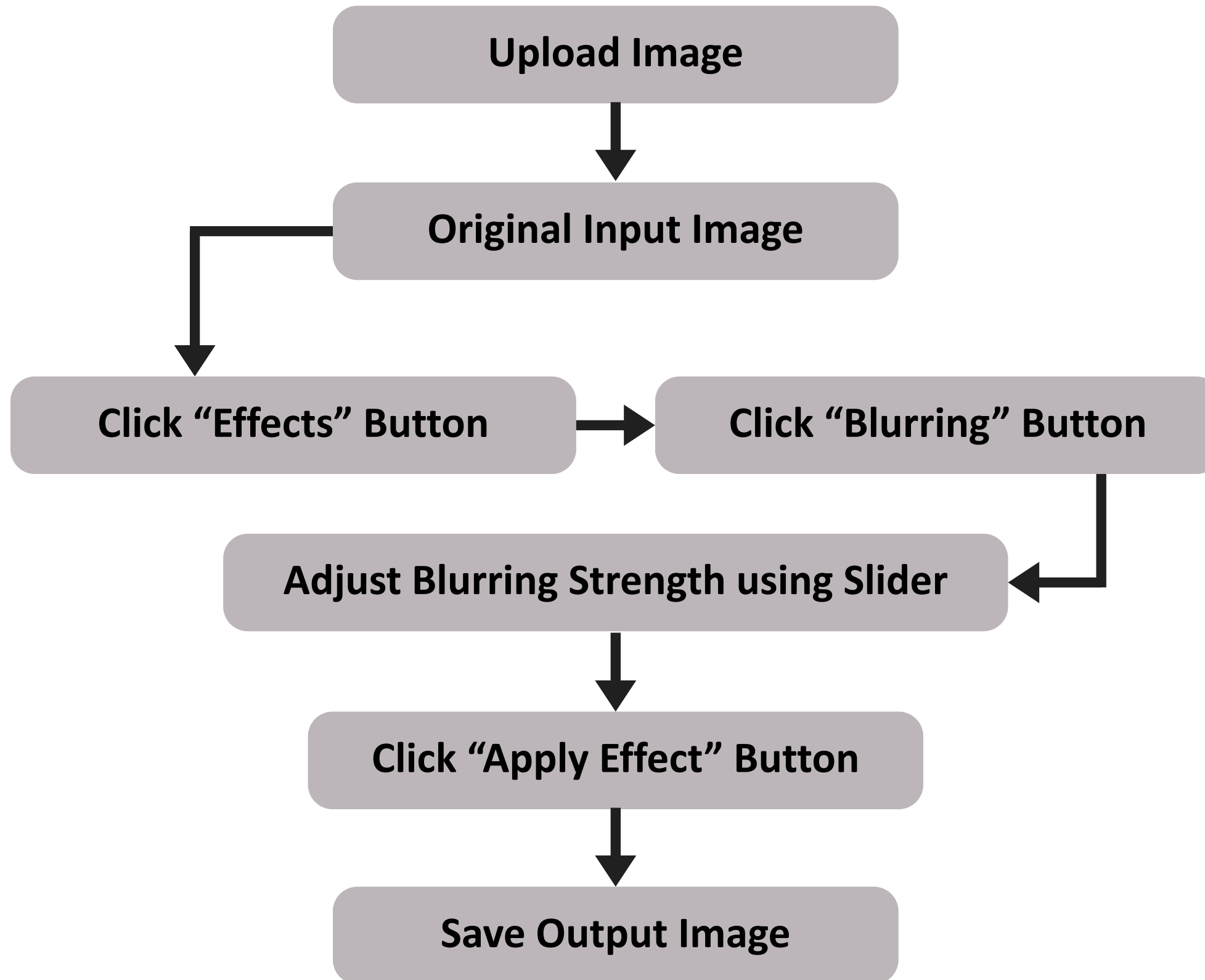
Sample Input and Output Process (Grayscale)



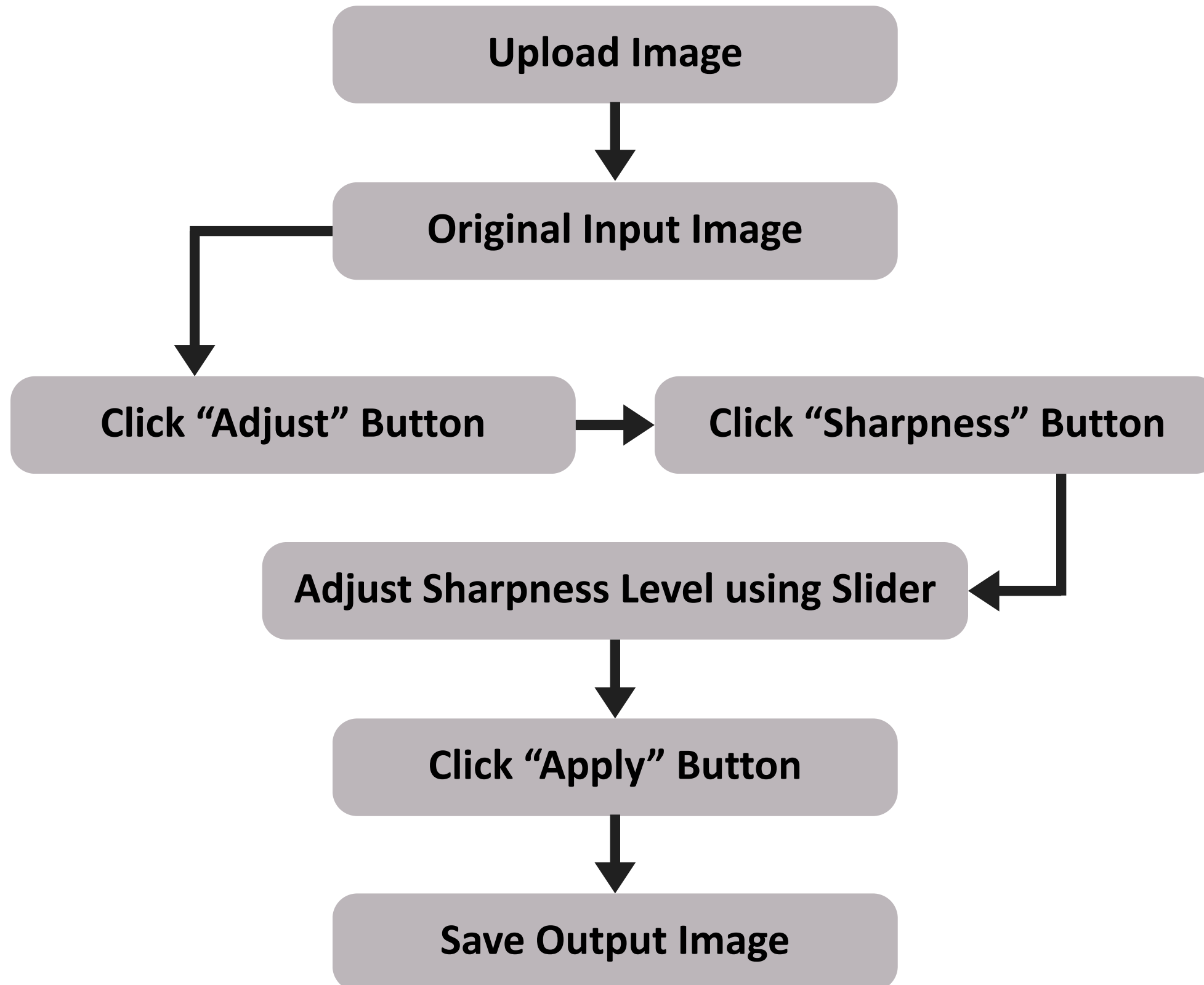
Sample Input and Output Process (Halftone)



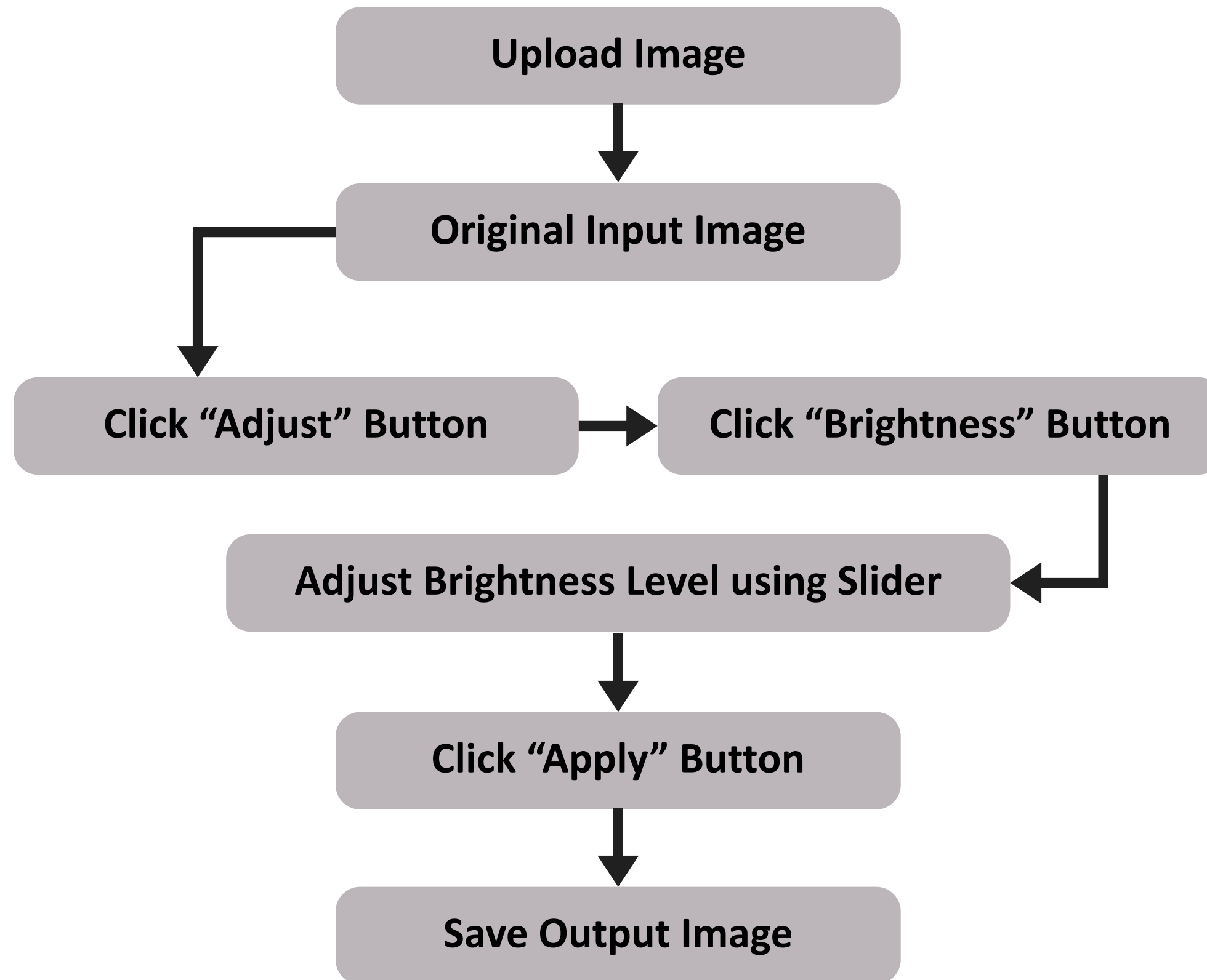
Sample Input and Output Process (Blurring)



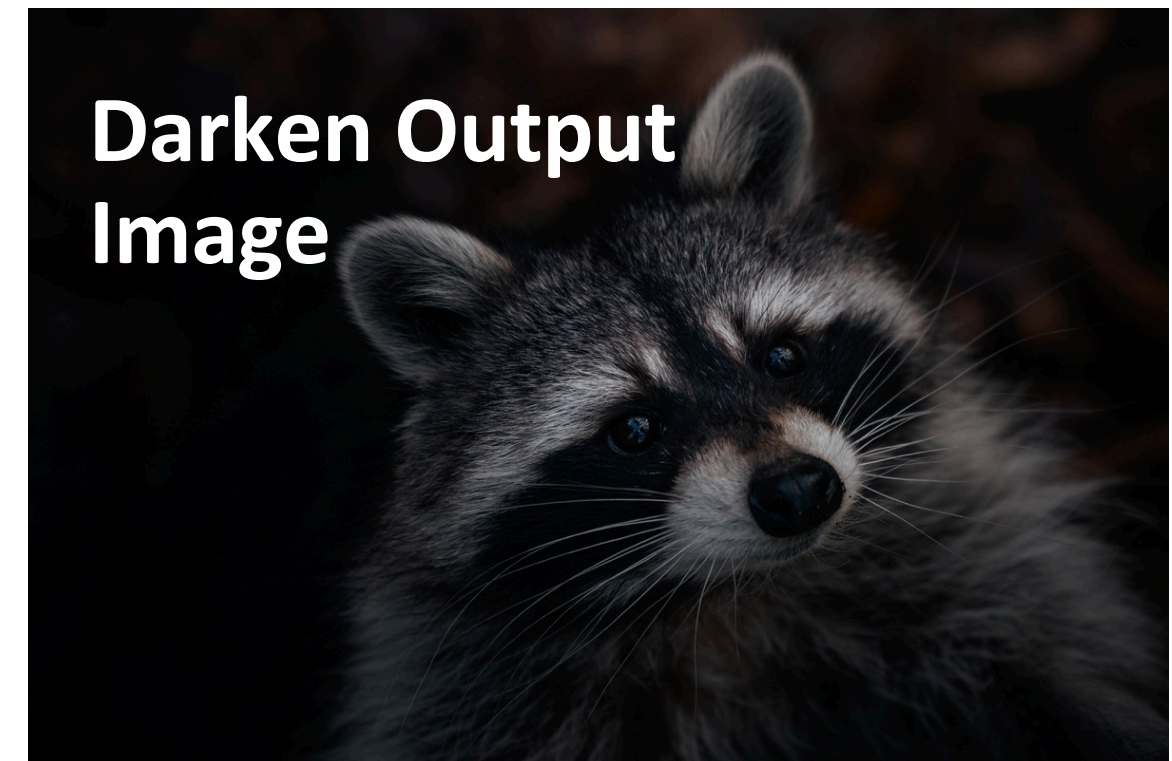
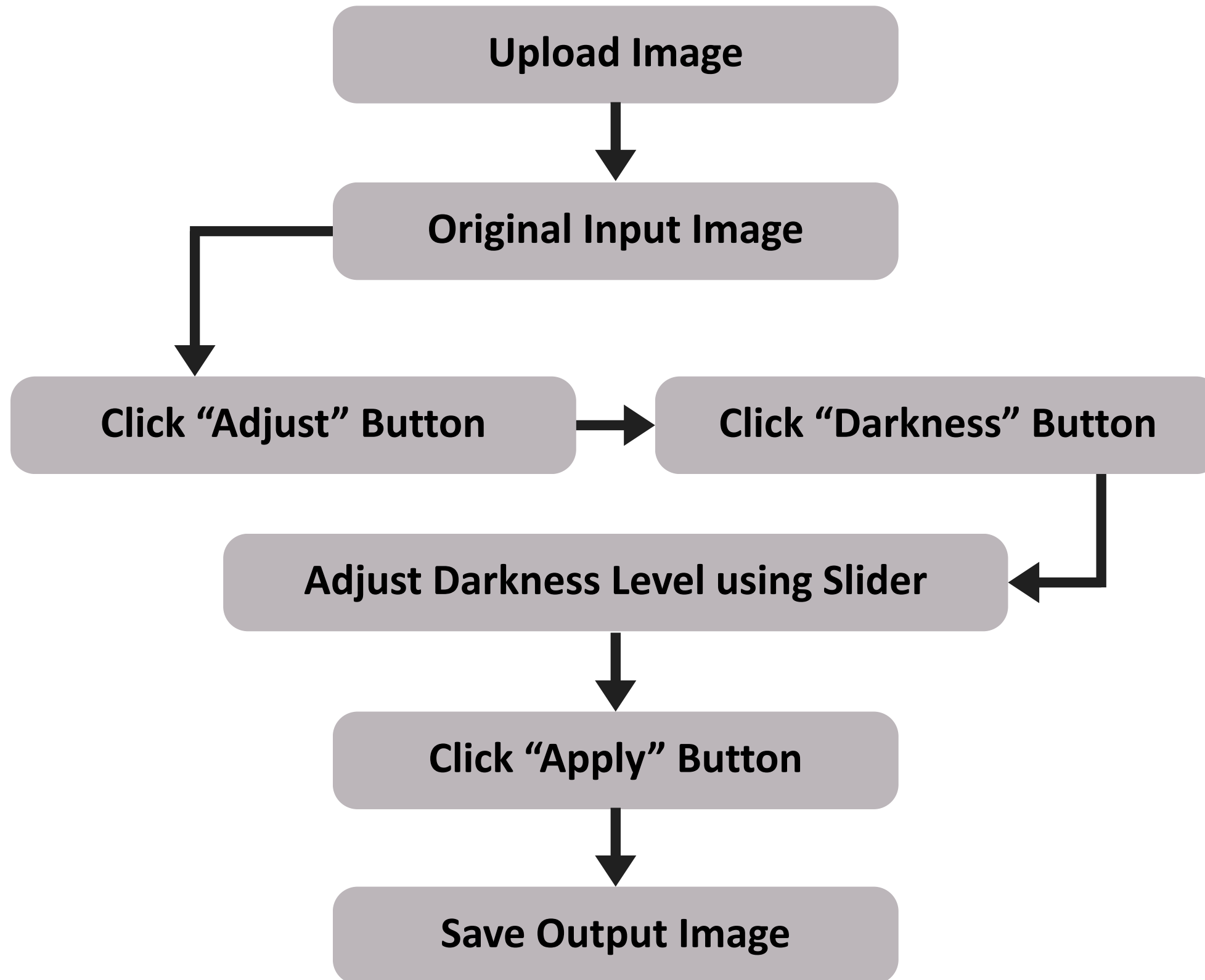
Sample Input and Output Process (Sharpness)



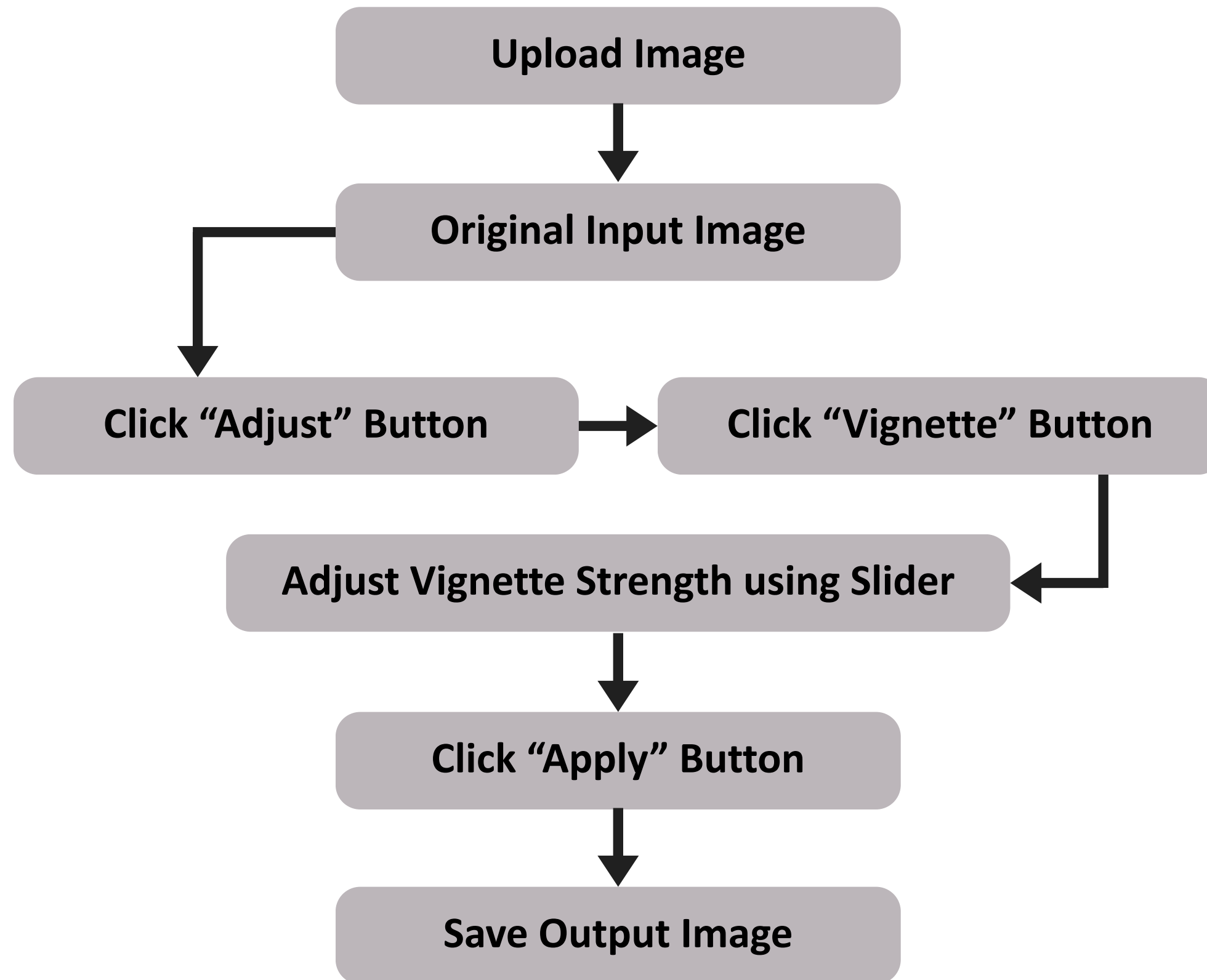
Sample Input and Output Process (Brightness)



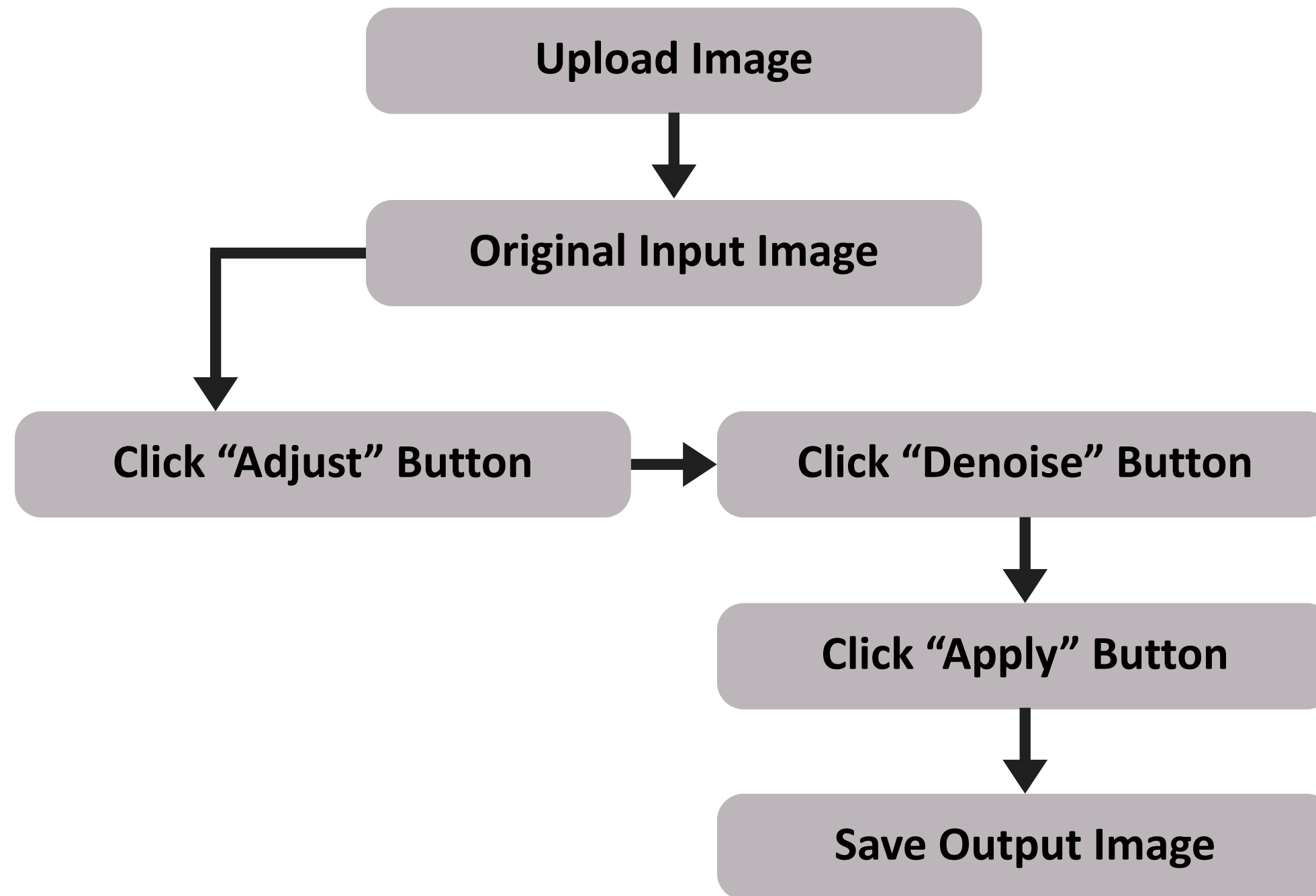
Sample Input and Output Process (Darkness)



Sample Input and Output Process (Vignette)



Sample Input and Output Process (Denoise)



Conclusion

In conclusion, this project is aim to implement a functional image processing application that demonstrates key concepts learned in this course. By integrating image enhancement, filtering, cropping and region based manipulation features, the proposed application will showcase the practical use of image processing techniques. This project is not only expected to enhance our technique skills in image processing and Python programming but also encourages teamwork, workload distribution and systematic documentation. Overall, the developed image processing application serves as a strong foundation for more advanced image editing system and provides valuable hands-on experience in designing real-world image processing solution.